Automated Detection of Satellite Trails in Astronomical Images Using Deep Learning

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MPhil Data Intensive Science

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1 Background and Motivation

Satellites reflect sunlight as they cross the sky, producing bright streaks in long-exposure telescope images. Pixels crossed by a trail can no longer be treated as reliable astronomical measurements. In the case, a whole image can be rejected, but this discards a large amount of otherwise useful data. Identifying the affected pixels instead allows them to be masked while preserving the remainder of the observation [1].

The scale of this problem is increasing rapidly. Between mid 2019 and early 2026, the number of human-made satellites in orbit rose from approximately 2000 to more than 14,000. Filings now cover more than 1.7 million planned satellites across 33 large constellations [2]. The expected growth threatens both ground-based [1] and space-based astronomy [3].

Stoppa et al. [4] addresses this problem with an automated pipeline developed using MeerLICHT observations. It divides each image into patches and applies a U-Net [5], a convolutional neural network [6] architecture, that assigns each pixel to a trail or the background. The patches are reassembled and a probabilistic Hough transform [7] fills gaps in trails and removes some non-trail artefacts. Figure 1 shows an image and its labelled mask from the released dataset [8].

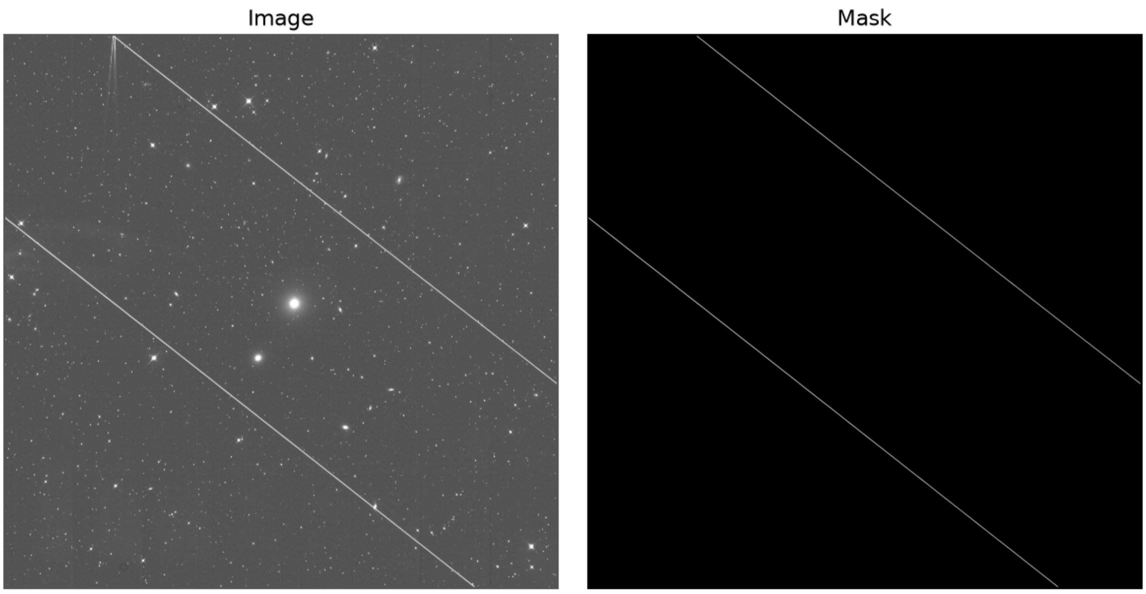

Figure 1: An example MeerLICHT image and its labelled satellite-trail mask from the ASTA dataset.

This project reproduces the central ASTA pipeline using the 178 released image-mask pairs. It also adds a lightweight classifier gate intended to improve processing speed and reduce false detections, and explores an Attention U-Net as an alternative segmentation model. However, training and evaluation of the Attention U-Net could not be completed within the available computational resources.

2 Methods

The pipeline patches the input image into non-overlapping 528×528 pixel sections. It then applies an optional convolutional classifier to filter patches before a U-Net then assigns each pixel to trail or background. Finally, a probabilistic Hough transform-based, geometric post-processing stage is applied to the full field predicted mask to reconnect gaps and remove non-trail artefacts such as star diffraction spikes. Because trail pixels are rare, training used a weighted sampler to oversample positive patches, combined with rotation, flip, and shift augmentation on trail-containing patches only. Operating thresholds for both models were selected from a sweep over validation data, balancing precision against recall before fixing them for test evaluation.

3 Results

The raw U-Net achieved a precision of 0.880, meaning that most pixels identified as trail were correctly labelled. Its recall was 0.914, meaning that it recovered most labelled trail pixels. Its IoU was 0.813. This measures the overlap between the predicted and labelled trail regions, with 1 representing exact agreement.

The classifier recovered 0.975 of trail-containing patches. Applying it as a gate raised segmentation precision to 0.889 and IoU to 0.820, while recall decreased slightly to 0.913. It reduced the false-positive rate from 7.68×10^{-5} to 7.02×10^{-5}, while marginally increasing the false-negative rate from 0.086 to 0.087. On the benchmark image, gating reduced segmentation runtime on CPU from 108.63 ± 2.04 seconds to 24.78 ± 1.24 seconds. Total A100 GPU runtime fell from 1.44 ± 0.01 seconds to 0.86 ± 0.02 seconds.

Geometric post-processing increased the recovery of trail pixels by reconnecting gaps, as illustrated in Figure 2. With the raw U-Net, the validation-selected settings reduced the false-negative rate, the fraction of labelled trail pixels left undetected, from 0.086 to 0.051. Recall rose to 0.949, although precision fell to 0.847 because additional background pixels were included. The resulting overlap score was 0.809. The classifier-gated equivalent produced a precision of 0.848, recall of 0.947, and overlap score of 0.810.

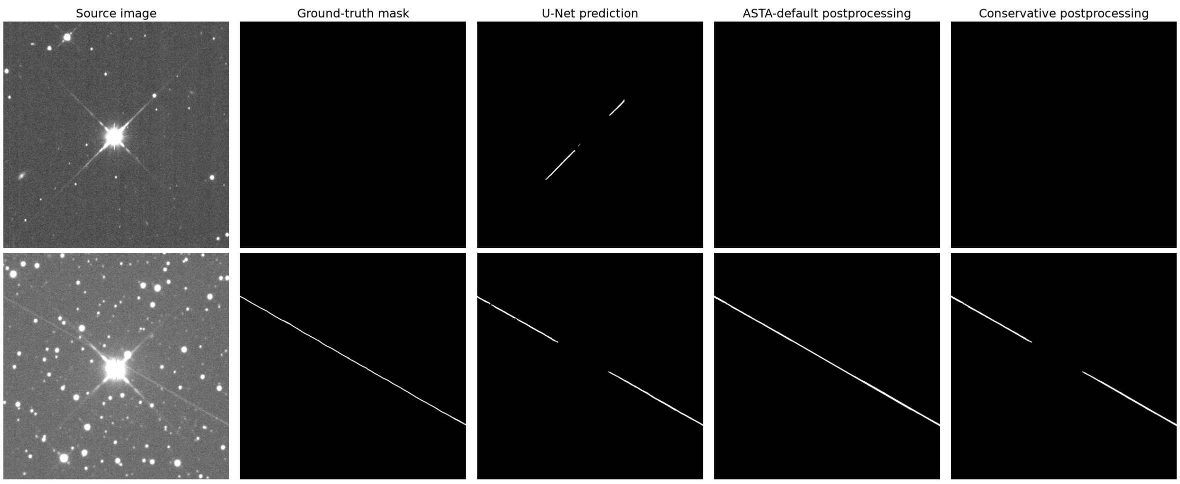

Figure 2: Examples of post-processing removing a stellar artefact and filling a gap where a star crosses a trail. Note that the more conservative postprocessing parameters fail to close this gap.

Figure 3 shows how the classifier can also prevent part of a stellar artefact from reaching the U-Net, allowing post-processing to remove the remainder.

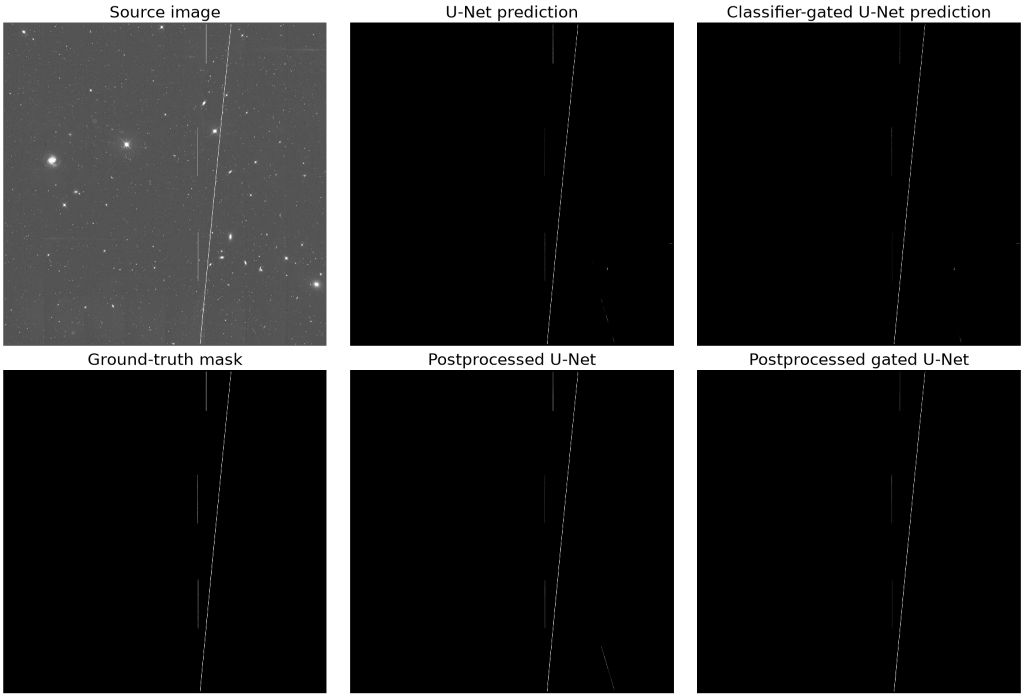

Figure 3: An example where classifier gating helps suppress a stellar artefact that the U-Net identifies as a trail.

4 Conclusions

This work reproduces the main qualitative result reported by ASTA. U-Net segmentation can locate satellite trails accurately, and geometric refinement can recover sections interrupted by stars, detector features, or patch boundaries. The numerical reproduction is not exact. ASTA reports approximately 0.94 precision and 0.94 recall, while the raw U-Net in this work achieved 0.880 and 0.914 respectively. ASTA also reports that refinement reduced the false-negative rate from 0.0698 to 0.0338, a relative reduction of approximately 51%. Using ASTA's released settings here reduced it from 0.086 to 0.046, a reduction of approximately 46%, and raised recall from 0.914 to 0.954.

The remaining gap is plausibly linked to data and evaluation differences. This reproduction used released 8-bit PNG images because the original higher-dynamic-range FITS data were unavailable. Its held-out test set was not involved in selecting model settings or thresholds. The results should therefore be viewed as a partial numerical reproduction and extension rather than an exact replication.

The classifier gate is the clearest new practical contribution. It improved raw overlap, retained almost all pixel-level recall, and substantially reduced CPU processing time. Post-processing presents a choice. It leaves fewer contaminated pixels undetected, but masks more clean pixels. The preferred configuration depends on the cost of missed contamination compared with losing unaffected data.

As satellites become more prolific, recovering usable data from contaminated observations will become increasingly important for ground-based astronomy. This will require highly accurate detection methods, and machine learning pipelines, such as the one explored here, are likely to play a growing role in meeting that need.

5 Next Steps

Future work should complete training and evaluation of the implemented Attention U-Net and test whether a larger segmentation model better separates trails from other linear artefacts. A stronger classifier could also be considered if its high recall is preserved. Evaluation on independent observations and synthetic trails would establish whether the models generalise beyond the limited MeerLICHT dataset. Access to the original FITS images would allow the effect of image representation on the performance gap to be measured directly.

References

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A Note on the Use of Autogeneration Tools

I have used OpenAI Codex to help me with the following tasks in completing this report:

- Formatting the LaTeXcode for the document
- Drafting initial text structure from bullet point outlines written by me, followed by my review and revision
- Final proofreading for accuracy and grammar
- Writing code to generate plots included within the report